

AP Calc Lesson 2.2 Homework

Name _____

1. The function f passes through the point $(-2, 5)$ and satisfies $f'(-2) = \frac{1}{3}$. Write an equation for the line tangent to the graph of f at $x = -2$.

2. Let function f be given by $f(x) = 5x^2 - 2x$.
 - a. Use the limit definition of the derivative to find an equation for $f'(x)$.

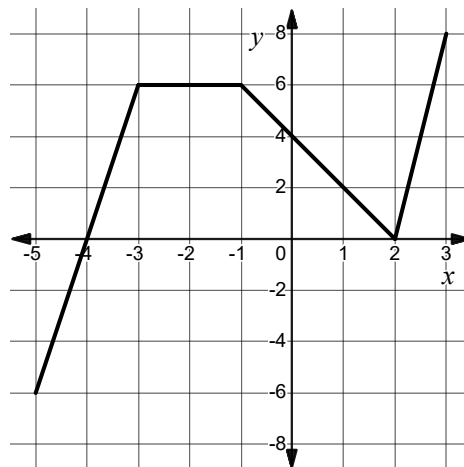
 - b. Find $f'(6)$.

 - c. Explain the meaning of your answer to part b.

3. The line tangent to the graph of function f at $x = 4$ has the equation $y = 5x - 2$.
 - a. Find $f(4)$. Explain how you found your answer.

 - b. Find $f'(4)$. Explain how you found your answer.

4. The graph of f is defined on the closed interval $[-5, 3]$ and is shown in the given graph. Use the same axes provided to graph f' , the derivative of f .



5. The graph of f' , the derivative of function f , is shown.

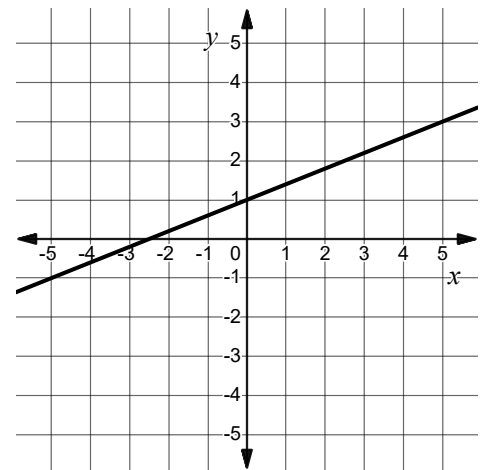
a. Find $f'(5)$.

b. Find an equation for $f'(x)$.

c. It is known

that $f'(10) = \lim_{h \rightarrow 0} \frac{(\frac{1}{5}(10+h)^2 + (10+h)) - 30}{h} = 5$. Write an equation for the line tangent to f at $x = 10$.

d. Find an equation for $f(x)$.



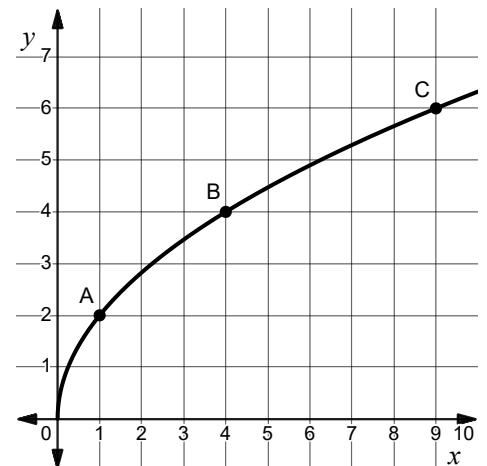
6. The graph of f' , the derivative of f , is shown.

a. What does Point A tell you about the graph of f ? Be specific.

b. Estimate the slope of f at $x = 7$.

c. Is the graph of f steeper at $x = 1$ or $x = 9$? How do you know?

d. Find the slope of the line normal to f at $x = 4$.

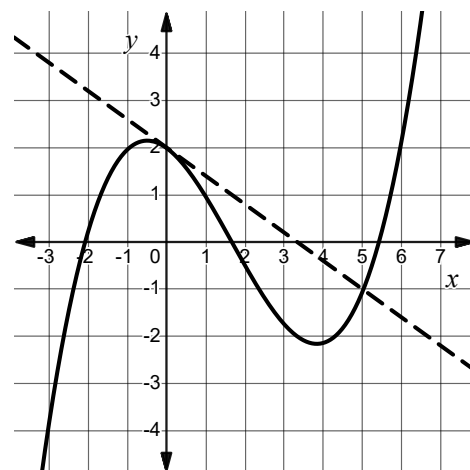


7. The graph of the function f is shown along with the line tangent to the graph of f at $x = 0$. For each of the following, find the value or explain why it cannot be found using the given information.

a. $f(0)$

b. $f'(0)$

c. $f(5)$



8. The line $y = b(x + 2)$ is tangent to the graph of function f at $x = -1$. The line $y = 2b(x + 1) + 1$ is tangent to the graph of f at $x = 0$. The lines tangent to f at $x = -1$ and $x = 0$ intersect when $x = -\frac{1}{2}$. Find $f'(-1)$ and $f'(0)$.

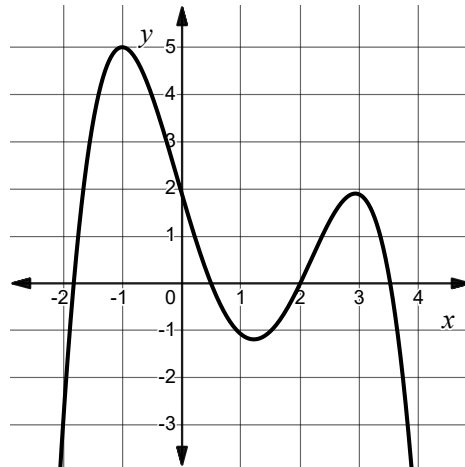
9. The graph of a function $y = f(x)$ is shown. Which of the following is true?

A) $f'(-1) < f'(0) < f'(2)$

B) $f'(2) < f'(0) < f'(-1)$

C) $f'(0) < f'(-1) < f'(2)$

D) $f'(-1) < f'(2) < f'(0)$



10. The line $y = L(x)$ is tangent to the graph of f at the point $(-4, 3)$ and $L(2) = 6$.

Evaluate $\lim_{x \rightarrow -4} \frac{f(x) - f(-4)}{x - (-4)}$.

A) -2

B) $\frac{1}{2}$

C) 2

D) undefined